

HEV Battery Management System (BMS)

Major Project

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Introduction

- ▶ Hybrid powertrains demand intelligent energy management and battery health monitoring.

BMS Architecture

- ▶ Battery Monitoring
- ▶ Cell Balancing
- ▶ Thermal Management
- ▶ Circuit Protection

Project Objectives

- ▶ Monitor SOC
- ▶ Analyze Performance
- ▶ Improve Reliability
- ▶ Energy Study

EV Energy Management Dataset

- ▶ Data Source
- ▶ Parameters (SOC & SOH)
- ▶ Utility and performance comparison

Simulation Methodology

- ▶ Data Collection
- ▶ GitHub Integration
- ▶ Python Analysis
- ▶ SOC & SOH Study
- ▶ Evaluation

Tools & Technologies

- ▶ Python
- ▶ GitHub
- ▶ Data Analysis Techniques
- ▶ Battery Management Concepts

GitHub Repository & Implementation

- ▶ Repository management, code hosting, dataset organization and documentation.

Result Validation Through SOC Monitoring

- ▶ Parameter Analysis
- ▶ SOC Accuracy
- ▶ Safety Evaluation
- ▶ Efficiency Gains

Benefits of BMS

- ▶ Life Extension
- ▶ System Efficiency
- ▶ Safety Assurance

Future Scope

- ▶ AI-Based Prediction
- ▶ Smart Charging (V2G)
- ▶ Real-Time Monitoring
- ▶ Advanced Thermal Systems

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Thank You

- ▶ Questions & Discussion
- ▶ Presented By: Poonam